

MAM · Continuous Random Variables and the Normal Distribution — Patterns and Solution Approaches

Basis: marking keys for 32 past exam questions from 2016–2025 + exam reports. Original questions in [topics/06_Continuous_RV_Normal.pdf](#). Numerical examples in this document have been checked with Python (scipy/sympy). 2016SA-Q12 refers to the 2016 official sample paper (not among the 32 questions; original paper in [papers/2016Sample_*](#)). Review status: has undergone one round of independent review and was revised according to feedback (see [Review_log.md](#)).

Marking panel reminders

- **Distinguish between pdf and cumulative distribution function** (2023 report: confusing pdf and cdf is very common).
- When finding k or proving k , you must first write "total area = 1" and then solve; you cannot just write the result (2023A-Q11(a) has 1 mark for "area under the curve must equal 1"; 2025F-Q4(a) key: no mark for simply writing the value of k without justification).
- You must be able to **name the distribution** (2021 report: in CF continuous probability questions, the distribution name is a source of lost marks; the key answer for 2021F-Q2(a) is "continuous uniform distribution").
- Expected values must be converted to the units or time in context: hours to minutes; $E(B) = 3.8$ h after noon written as 3:48 pm (2023A-Q11(c)). 2017A-Q11 and 2023A-Q11 each have 1 mark specifically for the conversion. The units of variance are squared units (e.g. L^2), and these must also be written (2022A-Q8(a) key: units for both expected value and variance).
- Follow the specified precision exactly when the question specifies it (three decimal places, exact, integer) (e.g. 2022A-Q12(a) has 1 mark for rounds to three decimal places); when none is specified, keep at least 4 decimal places (2020 report), and do not use only 2 decimal places in intermediate steps. For percentage questions, convert to a percentage.

Pattern 1: Relative frequency histogram → probability (CF)

Approach: Probability = sum of the relative frequencies of the corresponding bars; when an interval endpoint falls in the middle of a bar, **use linear interpolation assuming a uniform distribution** (this is how the 2023F-Q3(c) key does it). Conditional probability = intersection ÷ condition. All three days on time = p^3 (assuming independence).

- For a **frequency** histogram, first divide by the total number to convert to probability (2023F-Q3: 20 solar farms).
- Cumulative probability tables $P(W \leq w)$ and individual probabilities must be distinguished (specifically noted in the CF section of the 2023 report).
- Expected value estimate: $E \approx \sum \text{class midpoint} \times \text{probability}$ (2023F-Q3(c)(ii), 2024A-Q13(e)).
- Linear interpolation is valid only if the question states "assuming ... uniformly distributed within each interval".

Past exam questions: 2017F-Q1, 2020F-Q4(a)(b), 2023F-Q3, 2024A-Q13(e).

Pattern 2: Basic pdf operations ★

The four core formulas

- Total area: $\int f(x)dx = 1$ (used to find k or a);
- Probability: $P(a < X < b) = \int_a^b f(x)dx$;
- Expected value: $E(X) = \int x f(x)dx$; variance: $\text{Var}(X) = \int x^2 f(x)dx - \mu^2$, or $\int (x - \mu)^2 f(x)dx$;
- Median M : $\int_{\text{lower bound}}^M f(x)dx = 0.5$ (same for quantiles).

- Given the cumulative distribution function $F(x) = P(X \leq x)$: $P(a < X \leq b) = F(b) - F(a)$; pdf = $F'(x)$ (2023A-Q11(d)(e): the key awards 1 mark each for "express as $P(A \leq 4) - P(A \leq 3)$ " and "find the pdf").

CF techniques

- When the pdf consists of straight-line segments, use triangle/trapezium areas instead of integration (accepted in the keys for 2019F-Q3, 2022F-Q3 and 2025F-Q4).
- Piecewise pdf: integrate each segment separately; when finding k for the second part, set the area of the second part = $1 - \text{area of the first part}$ (2020F-Q4(d)).
- Example (2025F-Q1): $f(x) = 1/(x \ln 5)$, $1 \leq x \leq 5$. $E(X) = \int_1^5 1/\ln 5 \, dx = 4/\ln 5$; median: $\ln M / \ln 5 = 1/2 \Rightarrow M = \sqrt{5}$.

CA techniques: Write the integral expression (1 mark), then let the CAS calculate it; when finding c such that $P(X > c) = 0.01$, write the equation $\int_c^{\infty} f = 0.01$ (2022A-Q8(b), units 10 000 L, finally convert to the nearest litre).

Marking points: ✓ correct integral expression (including limits) ✓ calculation ✓ unit or time conversion.

Past exam questions: 2017A-Q11, 2018A-Q10, 2019F-Q3, 2020F-Q4(c)(d), 2022A-Q8, 2022F-Q6(b), 2023A-Q11, 2025A-Q13(c)(d), 2025F-Q1, 2025F-Q4.

Pattern 3: Uniform distribution

Formulas: $X \sim U(a, b)$: $f(x) = 1/(b - a)$ ($a \leq x \leq b$; 0 elsewhere); $E(X) = (a + b)/2$; $\text{Var}(X) = (b - a)^2/12$. **Note:** The formula sheet only lists $\text{Var}(X) = \int (x - \mu)^2 p(x) dx$, and **does not** list $(b - a)^2/12$. The key awards marks for "write the pdf and its domain" + "write the integral expression for variance" + "calculate the result" (2024F-Q4(a), 2019F-Q6(e), 2016A-Q16(b)). So $(b - a)^2/12$ is only for checking; in a formal answer you must write the integral.

Common variations

- Given $E(X) = 6$ and maximum 9 $\Rightarrow$ minimum 3 $\Rightarrow f(x) = 1/6$, $3 \leq x \leq 9 \Rightarrow \text{Var} = \int_3^9 (x - 6)^2 \cdot (1/6) \, dx = 3$ (2024F-Q4(a); can be checked with $6^2/12 = 3$).
- Conditional probability: $P(X < -0.35 \mid X < 0) = 0.15/0.5$ (2019F-Q6(c)).
- "square is less than 0.09" $\Rightarrow -0.3 < X < 0.3$ (2019F-Q6(d), the key has 1 mark for "express the required probability in terms of X ").
- "exactly 2 out of 3 days..." $\Rightarrow$ first find the single-day probability, then use $\text{Bin}(3, p)$ (2025F-Q4(c)).
- "What is the latest time to leave so that the probability of being late is no more than 4%?" $\Rightarrow$ cut off 4% of the area from the right end of the distribution (2021F-Q2(d)).

Past exam questions: 2016A-Q16, 2017F-Q2, 2019F-Q6, 2021F-Q2, 2024F-Q4(a), 2023A-Q13(a).

Pattern 4: Normal distribution — forward probability and inverse quantile calculations

Approach

- Write the probability expression: $P(X > 3.7)$, $P(395 < X < 405)$ (most keys award 1 mark separately, so always write it first).
- Use CAS normCdf to calculate, rounding to the required number of decimal places or converting to a percentage.
- Inverse: "heaviest 0.05%" $\Rightarrow P(X > m) = 0.0005 \Rightarrow \text{invNorm}(0.9995)$; "top 1%" likewise.
- Conditional probability: $P(X < 75 \mid X > 67) = P(67 < X < 75) / P(X > 67)$ (2020A-Q8(b), the key has 1 mark for "recognising that it is restricted to the jumbo range").
- Unit conversion: $Y = aX + b \Rightarrow \mu_Y = a\mu + b$, $\sigma_Y = |a|\sigma$, $\text{Var}(Y) = a^2\text{Var}(X)$ ($^{\circ}\text{C} \rightarrow ^{\circ}\text{F}$, km $\rightarrow$ mile; 2024A-Q13(c) asks for the variance). Inverse: given μ_Y , σ_Y , set up equations to find a , b (2025A-Q10(f)).

Past exam questions: 2016SA-Q12 (sample paper), 2017A-Q19, 2018A-Q12(a), 2019A-Q11, 2020A-Q8, 2021A-Q8, 2022A-Q12(a), 2024A-Q13, 2025A-Q10.

Pattern 5: Finding μ and σ from two probabilities ★

Recognition: Given two tail probabilities, find the mean and standard deviation. For example, 2016A-Q18: passengers wait for less than 55 minutes, 5% of the time, while there is a 13% chance that the waiting times will be greater than 100 minutes.

Fixed three steps (the key's marking points)

1. Find the two z-values separately (1 mark separately): $z_1 = \text{invNorm}(0.05) = -1.6449$, $z_2 = \text{invNorm}(0.87) = 1.1264$.
2. Set up the system of equations: $\mu - 1.6449\sigma = 55$, $\mu + 1.1264\sigma = 100$.
3. Solve to get $\sigma \approx 16.24$, $\mu \approx 81.71$ (2016A-Q18).

Another example (2023A-Q12): $P(X \leq 150) = 0.0228 \Rightarrow z = -2$, $P(X \geq 165) = 0.1587 \Rightarrow z = 1 \Rightarrow \sigma = 5$, $\mu = 160$ (0.0228 and 0.1587 are exactly the precise tail probabilities for $z = -2$ and $z = 1$, so z can be taken as an integer). When only the mean is known: one z-value is enough (2024A-Q13(a), 2019A-Q11(d)).

Past exam questions: 2016A-Q18, 2019A-Q11(d), 2023A-Q12(a), 2024A-Q13(a), 2025A-Q10(e).

Pattern 6: The 68–95–99.7 rule (CF)

Approach: Convert values into "mean $\pm k$ standard deviations": $163 \pm 3 \times 7 = 142$ to $184 \Rightarrow$ about 99.7%; $170 = \mu + \sigma \Rightarrow$ about 16% are greater than it; the shortest 2.5% are below $\mu - 2\sigma$. **Graph question (2021F-Q6):** The flatter and wider the curve, the larger the standard deviation; whether $P(6 \leq X \leq 9) \geq 0.5$: the key's reasoning is "total area = 1, the shaded area is clearly less than half". It can also be argued as follows: $\mu = 7$, $\sigma \approx 3$, the interval width $3 = \sigma$, and the maximum probability for an interval of width σ is only $P(\mu - \sigma/2 < X < \mu + \sigma/2) \approx 0.38 < 0.5$. For a continuous distribution, $P(Y = 2) = 0$, so $P(Y \geq 2) > P(Y > 2)$ shows that Y cannot be normal, but it can be binomial.

Past exam questions: 2018F-Q2, 2021F-Q6.

Pattern 7: Determining whether a model is appropriate (explanation question)

Wording: "The normal distribution is not appropriate because the histogram is not symmetric / is skewed to the right / is bounded at 0, whereas a normal distribution is symmetric and unbounded." or "the model gives $P(X < 0) = 0.0013$, but weight cannot be negative" (2018A-Q12(g)). **Past exam questions:** 2018A-Q12(g), 2020A-Q16(c), 2024A-Q13(d), 2025A-Q13(a).